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IFAC PapersOnLine 56-2 (2023) 11815–11820

Warehouse Path Planning Using Low-order Bézier Curves with Minimum-Time Optimization ⋆

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Abstract:
In this paper, low-order Bézier motion primitives are proposed for motion planning in corridor-constrained spaces, which is illustrated on a warehouse application. The proposed primitives are easily computed for given requirements (initial and final position, orientation and curvature). A new parametrization of the motion primitives is provided which allows an intuitive geometric interpretation. The combined path constructed from these primitives has a minimal number of turns, as third-order Bézier curves are used. At the same time, the combined path has continuous curvature. A path planning approach is proposed that finds the optimal combination of primitives that results in a minimum travel time while satisfying constraints on velocity, acceleration, and free space. To illustrate the applicability of the approach, a comparison with a typical predefined warehouse path and an existing advanced optimisation planner is provided and evaluated.

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Keywords: path planning, motion primitives, Bezier curve, continuous curvature, minimal time, warehouse

1. INTRODUCTION

Modern warehouses (and other logistics systems) tend towards complete automation that enables quick delivery of the desired goods to a customer, eliminates manual physical work and even manual data entry and analysis. In order to achieve warehouse automation, several layers of control and information technology support have to be designed appropriately. Such systems are a vital part of logistics operations for major online retailers that frequently store millions of items and operate with large and variable daily order volumes. These businesses often adopt automated guided vehicles (AGVs) as a flexible alternative to perform and optimize various tasks: transportation, loading and unloading of goods, item stacking and retrieving, order picking and sorting, inventory tracking, and warehouse maintenance. The advantages of physical automation include increased warehouse capacity and efficiency, improved reliability, availability, and productivity (Azadeh et al., 2019; Vivaldini et al., 2010; Tao et al., 2021; Zamfirache et al., 2022). Although numerous sites have already been automated by AGVs or autonomous mobile robots (AMR) a 30 % growth is expected in the next five years (Fragapane et al., 2021).

Traditionally AGVs operate on predefined fixed routes which simplifies and lowers the complexity of path planning. However, the coordinated planning for a group of AGVs that result in a higher throughput at the less likely appearance of conflicts still is a challenging task (Digani et al., 2015). To further increase the flexibility of AGVs or AMRs can move freely inside provided free space which is limited by corridors. Different planning algorithms can be applied to obtain paths. Typically, graph-search algorithms such as A*-based algorithms on a regular grid can be used where the path is a sequence of straight-line segments with discontinuities in the tangents and curvatures in the joints. For wheeled vehicles with kinematic constraints planning algorithms that use motion primitives are a more appropriate choice. Popular parametric and non-parametric motion primitives include clothoids (Gim et al., 2017), Bézier curves (Choi et al., 2010; Chen et al., 2014; Zdešar and Škrjanc, 2018; Klančar et al., 2021), B-splines, or higher order polynomials (Ghilardelli et al., 2014).

Approaches that use motion primitives for path planning include graph-based search algorithms and hybrid planners

⋆ This work was founded by the Slovenian Research Agency under Grants P2-0219 and L2-3168.

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10.1016/j.ifacol.2023.10.578

such as (Likhachev and Ferguson, 2009; Dolgov et al., 2010; Klančar et al., 2021). In large environments, randomized motion planners are prefered such as Rapid Exploring Random Tree (RRT*) and the kinodynamic RRT* (Li et al., 2016; Webb and van den Berg, 2013).

Additionally, vehicles need to plan their velocity profiles to be consistent with their dynamic constraints (maximal velocity and acceleration). Dynamic constraints are also imposed by the load which can dictate slower motions with moderate acceleration to avoid damage to the load. Therefore, also feasible velocity profile needs to be planned for existing paths. Because time efficiency is crucial in modern warehouses both path and velocity profiles are jointly planned to result in safe and time-minimal trajectories for transportation. Since trajectory planning is complex a path-velocity decomposition is typically applied, wherein the second step admissible velocity profiles are determined that meet time-related requirements such as minimising travel time or optimising driving comfort while taking into account dynamic constraints on a predefined path (Velenis and Tsiotras, 2008; Klančar et al., 2017; Loknar et al., 2021).

In this work, we focus on the problem of constructing the path of the automated vehicles inside the warehouse. The objective is to find a planar path that visits the goal points and provides the minimum driving time while respecting the forbidden areas and keeping the acceleration and velocity within bounds. Path planning is based on third-order Bézier motion primitives that can easily be combined to obtain continuity in tangency and curvature. An optimisation algorithm is proposed to obtain a minimal time trajectory that visits required stop locations and is located inside a safe corridor. The obtained trajectory is compared to a simple usual-appearance warehouse path and to the state-of-the-art race-track optimisation planner.

2. PATH PLANNING

Path planning starts by identifying the goal. In this context, this is done by specifying the initial and final pose of the AGV. We will also assume that the path should be continuous, and respects the standard kinematic limitations. In particular, the curvature of the path should be bounded. The upper bound is defined by the lowest achievable turning radius of the vehicle. Even if the vehicle can achieve an infinite curvature (it turns in a spot), lower-curvature paths usually result in a shorter driving time. Bounded-curvature paths can always be driven with such a velocity that the resulting tangential and radial accelerations can be kept below a given bound. The proposed approach will result not only in bounded-curvature paths but also in continuous-curvature paths. This is particularly important since it gives an additional benefit of bounded jerks (the third derivative of the position). High jerks result in similar problems as high acceleration, i.e. excessive stress and vibrations, driving discomfort etc.

2.1 Low-order Bézier curves as motion primitives

Bézier curves are parametric curves that are often used in scientific work and practical applications (Fitter et al., 2014; BiBi et al., 2019; Meng et al., 2021; Maqsood et al., 2020). A Bézier curve of the n-th order is defined as:

$$\mathbf{B}_n(\lambda) = \sum_{i=0}^{n} \mathbf{P}_i \binom{n}{i} \lambda^i (1-\lambda)^{n-i}, \quad \lambda \in [0,1], \qquad (1)$$

where $\mathbf{P}_i$ $(i = 0, 1, 2, \dots n)$ are the so-called control points. In this work, only planar curves in the (x, y) plane will be dealt with, so $\mathbf{B}_n(\lambda), \mathbf{P}_i (i = 0, 1, 2, \dots n) \in \mathbb{R}^2$. It is easy to show that the Bézier curve always starts and ends in the two extreme control points $\mathbf{P}_0$ and $\mathbf{P}_n$ while the initial and the final orientations are given by the direction of the vectors $(\mathbf{P}_1 - \mathbf{P}_0)$ and $(\mathbf{P}_n - \mathbf{P}_{n-1})$, respectively.

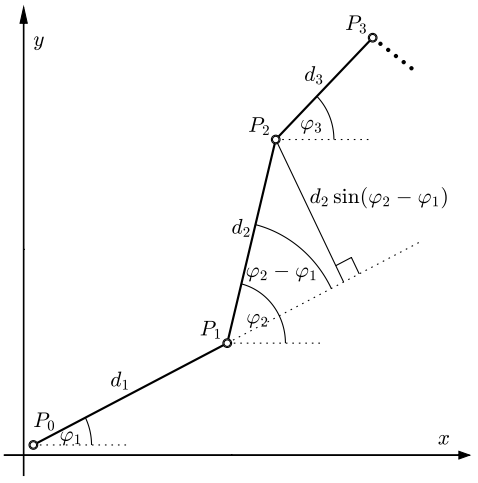


Fig. 1. Geometric relations among the control points of a Bézier curve: d_i represents the length of the line segment P_iP_{i+1} while φ_i is its orientation.

In this work, we will limit ourselves to the low-order Bézier curve motion primitives. Thus, we avoid very complex shapes of the motion primitives. To illustrate this, it could be shown that the sign of the curvature can change the sign as many as $(2n-4)$ times. So, to prevent these large oscillations of the curvature of motion primitives, we chose third-order motion primitives. In this case, the primitive is defined by four control points as illustrated in Fig. 1. In this work, the curvature in the motion-primitive endpoints will play a crucial role. The curvature of the parametric curve is given as:

$$\kappa = (x'y'' - y'x'')(x'^2 + y'^2)^{-\frac{3}{2}}, \qquad (2)$$

and consequently, the following limits can be obtained in the endpoints:

$$\begin{aligned} \kappa_0 &= \frac{2}{3} \frac{D \sin(\phi - \varphi_1) - d_3 \sin(\varphi_3 - \varphi_1)}{{d_1}^2} \\ \kappa_3 &= \frac{2}{3} \frac{D \sin(\varphi_3 - \phi) - d_1 \sin(\varphi_3 - \varphi_1)}{{d_3}^2} \end{aligned} \qquad (3)$$

where D and ϕ represent the Euclidean distance between P_0 and P_3, and the orientation of the line segment P_0P_3 (by orientation we mean the angle between the line segment and the positive direction of the x axis), respectively.

In this work, we will define the motion primitive by its endpoints, orientations in endpoints (φ_1 and φ_3), and curvatures in endpoints (κ_0 and κ_3). The problem of finding the motions primitive then transforms into the solution of the system of two quadratic equations with two unknowns (d_1 and d_3) that is obtained from Eq. (3):

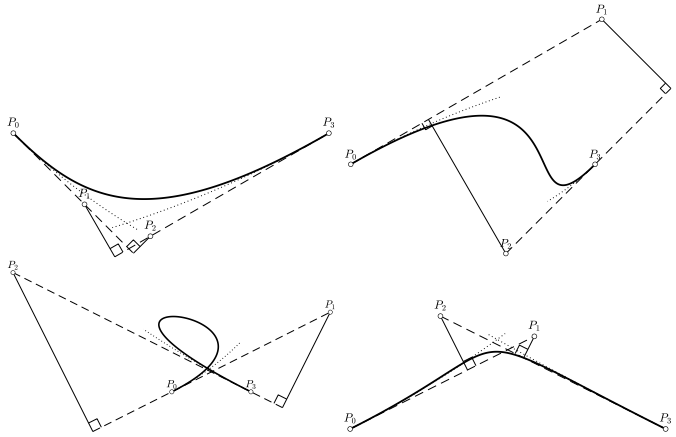


Fig. 2. Four basic motion primitives: C-shaped, S-shaped, loop, and V-shaped.

$$d_3 = \frac{D\sin(\phi-\varphi_1)}{\sin(\varphi_3-\varphi_1)} - \frac{3\kappa_0}{2\sin(\varphi_3-\varphi_1)}{d_1}^2$$
$$d_1 = \frac{D\sin(\varphi_3-\phi)}{\sin(\varphi_3-\varphi_1)} - \frac{3\kappa_3}{2\sin(\varphi_3-\varphi_1)}{d_3}^2 \quad (4)$$

It was shown (Blažič and Klančar, 2022) that a unique solution almost always exists. The solution does not exist if the given parameters in the endpoints are such that there does not exist a simple curve meeting those boundary conditions. If this is the case, the motion primitive should be broken into two segments. The solution (d_1 and d_3) of the system of equations (4) completely defines all the control points and the motion primitive itself. Four general types of motion primitives can be obtained as illustrated in Figure 2.

2.2 Construction of the path from motion primitives

In subsection 2.1, we have seen that 3rd order Bézier curves can serve as motion primitives that achieve desired position, orientation, and curvature in the endpoints. This subsection will provide the algorithm to compose the whole path from these motion primitives. It consists of the following steps:

(1) Identify the region where the path should be enclosed. Typically, it can be seen as a corridor which starts at the starting point and finishes at the ending point.
(2) Identify the intermediate points. They are placed where the "curvature" of the corridor reaches the local extrema. In a typical warehouse layout, the intermediate points are placed on the inner corners of the turns.
(3) Provide the curvatures and the orientations in the intermediate points.
(4) Solve the system of quadratic equations (4) for d_1 and d_3 that completely define the control points of the motion primitive as illustrated in Fig. 1 and the motion primitive itself as shown in (1).

It has to be stressed here that the orientations and the curvatures in the intermediate points in step 3 can be obtained by trial-and-error, using some ad-hoc estimation, or by optimisation. Usually, a good initial guess can be obtained by observing the geometric properties of the corridor. Later, these values can be updated to optimise some criterion as will be shown in the example in section 4.

2.3 Path obtained by receding horizon optimisation of curvature

For the purpose of comparison, the optimisation-based path planner in a racing track (Bonab and Emadi, 2019) idea is adopted and modified to warehouse problem. The receding horizon planner optimises path curvature inside provided corridor. The path is defined by a sequence of points driven by a vehicle with discrete vehicle kinematics

$$\begin{aligned} x(i+1) &= x(i) + \Delta s(i)\cos\left(\theta(i) + \frac{\Delta s(i)\kappa(i)}{2}\right) \\ y(i+1) &= y(i) + \Delta s(i)\sin\left(\theta(i) + \frac{\Delta s(i)\kappa(i)}{2}\right) \\ \theta(i+1) &= \theta(i) + s(i)\kappa(i) \end{aligned} \quad (5)$$

where $x(i)$, $y(i)$, $\theta(i)$ is the pose of the vehicle at the i-th point along the path and control variables are the travelled distance $\Delta s(i)$ and curvature $\kappa(i)$. $\Delta s(i) = \Delta s$ is chosen as small enough and constant. That results in the path being defined by the initial vehicle pose and the course of $\kappa(i)$. The optimal path is obtained by constrained optimisation which minimizes curvatures $\kappa(j)$ for the prediction horizon $j \in i, \cdots, i+h$ so that the resulting path remains in the corridor.

$$J(i) = \min_{\kappa(j)} \sum_{j=i}^{i+h} \kappa^2(j)$$
$$\text{subject to: stay inside provided corridor.} \quad (6)$$

The first element of $\kappa(j)$, $j \in i, \cdots, i+h$ is applied to obtain the next vehicle pose $(i+1)$ using kinematics (5) in receding horizon fashion.

2.4 Path planning using line segments and circular arcs

In this section, we will only provide a very simple algorithm that will be used solely for the purpose of comparison. The path is only composed of line segments and circular arcs. The algorithm consists of the following steps:

(1) Straight line segments are placed either along the centre of the corridor (when the vehicle does not stop on this segment) or along the border of the corridor (when there is a stopping point on this segment).
(2) Circular arcs are placed to make continuous transitions of the orientation when moving from one line segment to the next.

3. OPTIMAL VELOCITY PROFILE DETERMINATION

In this section, we will treat the problem of minimizing the driving time of a predefined path by taking into account the velocity and acceleration limitations. We will assume that the maximal velocity of the vehicle (v_{MAX}) is given. Also, the maximal tangential acceleration (a_{MAXt}) and maximal radial acceleration (a_{MAXr}) are known, and it is assumed that the overall acceleration is bounded by the ellipse:

$$\frac{a_t^2}{a_{MAXt}^2} + \frac{a_r^2}{a_{MAXr}^2} \leq 1 \quad (7)$$

where $a_t = \dot{v}$ is the tangential acceleration, and $a_r = \kappa v^2$ is the radial acceleration. The used vehicle limitations are met quite often in the literature (Velenis and Tsiotras, 2008; Klančar et al., 2017; Loknar et al., 2021) .

Here, the minimum driving time along the given path is sought. It is obtained by maximizing the velocity profile. In other words, the resulting profile gives the velocity of the vehicle with respect to time (or travelled distance) that cannot be increased at any point without violating either velocity or acceleration limitation. The whole algorithm is given by Lepetič et al. (2003); Klančar et al. (2017). The algorithm adopted in this paper is only slightly modified since we request a stop of the vehicle at some points. The input to the algorithm is the path that is arbitrarily parametrised. The curvature is given by Eq. (2). The sketch of the algorithm is as follows:

(1) Determine the initial points of the velocity profile determination where the velocity is known. They consist of:
   - Turning points. These are the extremal points of the absolute value of the curvature. In the turning points, the velocity achieves the minimum, and tangential acceleration is 0 while the radial acceleration is at the upper bound. Then, the velocity is known from $a_r = \kappa v^2$.
   - Stopping points. These are the points where the vehicle has to stop, e.g., it has to either collect or drop an object in the warehouse. The velocity is zero there.
   - Starting and final points. The velocity is also given in these points.

(2) The velocity around these initial points is obtained by integrating forward ($\dot{v} > 0$) and backward ($\dot{v} < 0$) while keeping the velocity high enough so either acceleration or velocity is on its boundary. In other words, the following differential equation is solved numerically until $|v| = v_{MAX}$ is achieved (the maximal value of the velocity is kept as a solution until $|v|$ drops below v_{MAX}):

$$\frac{(\dot{v})^2}{a_{MAXt}^2} + \frac{(\kappa v^2)^2}{a_{MAXr}^2} = 1 \tag{8}$$

with the given initial velocity in the initial points of integration. It has to be stressed here that the integration is done according to the parameter of the curve and not the actual time.

(3) The final velocity profile is obtained by minimizing all the obtained velocity profiles which guarantees that the limitations are never violated.

(4) The velocity profile is transformed to the time domain which means that the parameter of the curve is mapped to the actual time. This, in turn, gives the driving time of the whole curve.

4. WAREHOUSE PATH PLANNING CASE STUDY

In this section, the results of a simulation study will be shown where the paths for the AGVs in the warehouse are constructed. The layout of the warehouse is shown in Fig. 3. The dark grey regions represent the areas that the AGVs cannot enter (storage area with shelves, shipping area, staging area etc.). It has to be stressed here that the area of these patches is larger than the actual forbidden area so that the AGV can be represented as a point in the subsequent simulations.

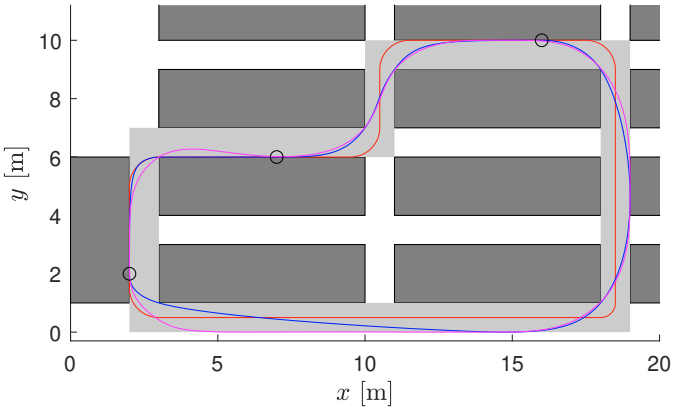


Fig. 3. The layout of the warehouse and the paths obtained by three algorithms given in subsections 2.2 (blue), 2.3 (magenta), and 2.4 (red).

The three black circles depict the locations where the AGV is to stop. The leftmost circle depicts the location of the pick-up/drop-off point. This point will serve as a starting and an ending point for all the paths. The other two points represent the locations where the items are to be collected or dropped off. All the circles should therefore lie on the path and the velocity in these points should be zero.

The light grey area represents the corridor that has to include all the paths. Here we do not treat the problem of finding this corridor. In the example studied, the corridor follows directly from the known stopping points.

4.1 The algorithm given in subsection 2.2

This is the main algorithm proposed in this work. The algorithm starts by identifying the intermediate points. These are found on the inside corner of each turn. Additionally, all three stopping points are also intermediate points so that they are guaranteed to appear along the resulting path. The orientations and curvatures in the intermediate points are also needed. In this case, they were searched for by optimization where the cost function consisted of the part that minimizes the integral of κ^2 along the path, and the second part was added to penalize the curves that leave the corridor. The resulting path tends to low curvature which is advantageous when searching for the shortest driving time.

Note that the driving time is not explicitly minimised during optimisation. This is done to make the comparison to the algorithm given in subsection 2.3 fair.

4.2 The algorithm given in subsection 2.3

For an approximately 47 m long corridor, the distance increment is set to $\Delta s = 0.23$ m thus the optimal path has around 200 points and the horizon is set to $h = 70$ points (35% of the total path length). The optimisation (6) is therefore computationally demanding as it includes a large number of parameters (e.g., curvatures in 70 points) that are optimised for every point of the path. The algorithm also needs some tweaking to force the path to approach

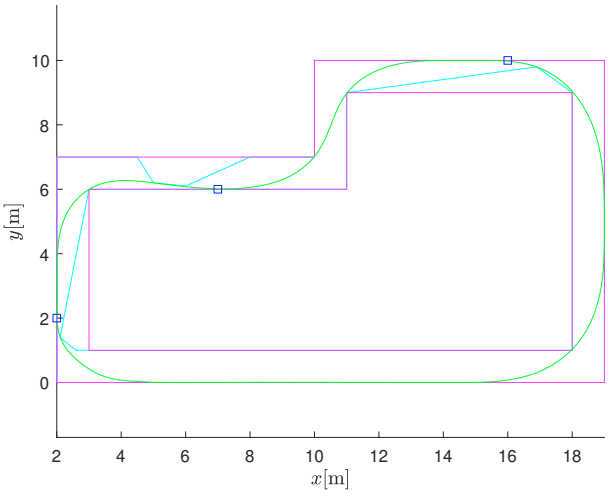


Fig. 4. Optimal path obtained by curvature optimisation (green line). The path is inside the corridor (pink line) where additional more restrictive corridor narrowing (cyan line) is introduced to approach defined stopping points (blue squares).

the stopping points (denoted by square signs in Fig. 4) although it is not possible to cross them exactly. This is done by more restrictive narrowing of the corridor (cyan lines in Fig. 4).

4.3 The algorithm given in subsection 2.4

This algorithm is given here only to provide a very simple reference and a solution that can be applied very easily. It is extremely computationally efficient and easy to program and implement. However, this solution suffers from the problem that the curvature is not continuous at the point where line segments meet circular arcs. As a result, there is a sudden change in radial acceleration during driving through these points. This is important because it introduces instantaneous changes in forces acting on moving objects.

4.4 Comparison of the results

The paths of the three algorithms are shown in Fig. 3. The velocity profiles are shown in Fig. 5 and the curvatures in Fig. 6. The driving times of the first path (36.4357 seconds) and the second path (36.4772 seconds) are just 0.04 seconds apart. The driving time of the third path is 41.7813 seconds which is more than 5 seconds slower than the other two algorithms achieve.

5. CONCLUSION

The subject of the paper is the development of the algorithm of path construction for AGVs in a warehouse. The problem of minimum time paths is studied taking into account velocity and acceleration constraints. Minimising the travel time of the AGVs leads to an increase in the throughput of the warehouse. However, the travel time is not explicitly minimised. Another criterion is used which requires less computation but gives a similar solution. This is done to allow a fair comparison with an existing advanced optimisation planner. Both planners give almost

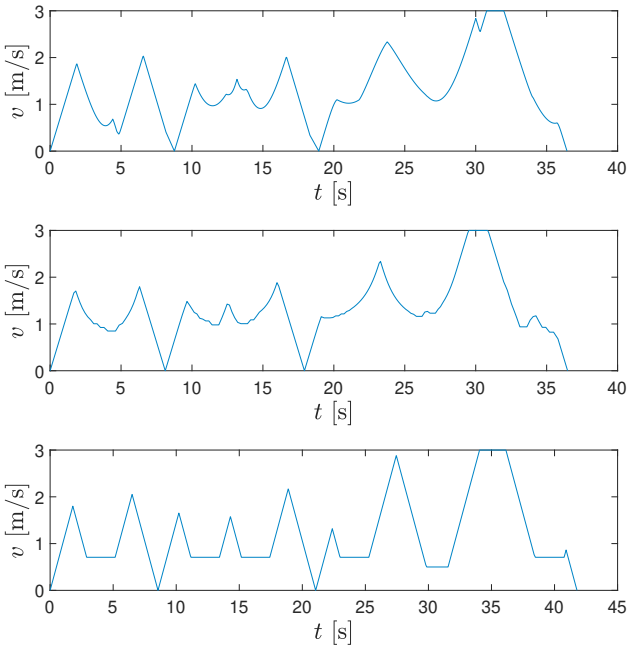


Fig. 5. The velocity profiles obtained by three algorithms given in subsections 2.2 (top), 2.3 (middle), and 2.4 (bottom).

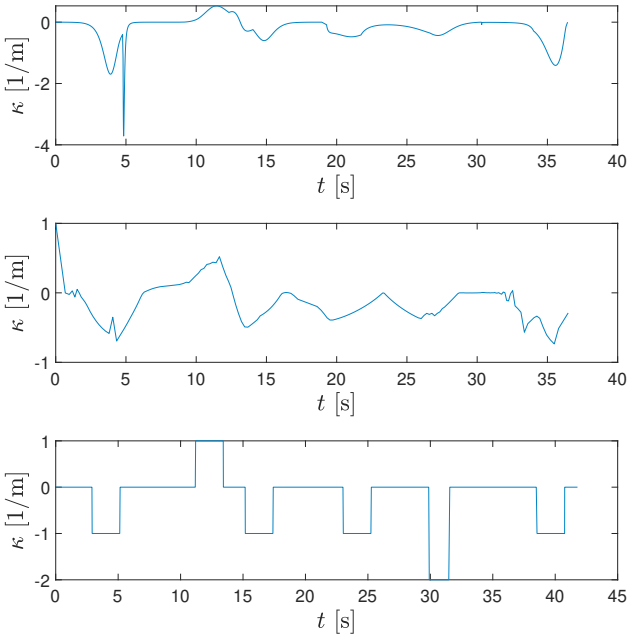


Fig. 6. The curvatures obtained by 3 algorithms given in subsections 2.2 (top), 2.3 (middle), and 2.4 (bottom).

the same travel time, although the proposed algorithm achieves a slightly shorter time, and is also less computationally demanding. The algorithm could easily be adapted to minimise the actual travel time. This would allow the shape of the curves to be optimised for different acceleration limits resulting from different loads on the AGVs. Future work could extend the algorithm to multi-vehicle path planning, where potential collisions could be checked and, if necessary, new segments added to avoid collisions.

REFERENCES

Azadeh, K., Koster, R.D., and Roy, D. (2019). Robotized and automated warehouse systems: Review and recent developments. *Transportation Science*, 53(4), 917–945. doi:10.1287/trsc.2018.0873.

BiBi, S., Abbas, M., Misro, M.Y., and Hu, G. (2019). A novel approach of hybrid trigonometric bézier curve to the modeling of symmetric revolutionary curves and symmetric rotation surfaces. *IEEE Access*, 7, 165779–165792. doi:10.1109/ACCESS.2019.2953496.

Blažič, S. and Klančar, G. (2022). Effective parametrization of low order bezier motion primitives for continuous-curvature path-planning applications. *Electronics*, 11(11). doi:10.3390/electronics11111709.

Bonab, S.A. and Emadi, A. (2019). Optimization-based path planning for an autonomous vehicle in a racing track. In *IECON 2019 - 45th Annual Conference of the IEEE Industrial Electronics Society*, volume 1, 3823–3828. doi:10.1109/IECON.2019.8926856.

Chen, C., He, Y., Bu, C., Han, J., , and Zhang, X. (2014). Quartic Bezier Curve based Trajectory Generation for Autonomous Vehicles with Curvature and Velocity Constraints. In *IEEE International Conference on Robotics and Automation (ICRA)*, 6108–6113. Hong Kong.

Choi, J.W., Curry, R., and Elkaim, G. (2010). *Machine Learning and Systems Engineering*, chapter Piecewise Bezier Curves Path Planning with Continuous Curvature Constraint for Autonomous Driving, 31–45. Lecture Notes in Electrical Engineering 68. Springer Science+Business Media.

Digani, V., Sabattini, L., Secchi, C., and Fantuzzi, C. (2015). Ensemble coordination approach in multi-agv systems applied to industrial warehouses. *IEEE Transactions on Automation Science and Engineering*, 12(3), 922–934. doi:10.1109/TASE.2015.2446614.

Dolgov, D., Thrun, S., Montemerlo, M., and Diebel, J. (2010). Path planning for autonomous vehicles in unknown semi-structured environments. *The International Journal of Robotics Research*, 29(5), 485–501.

Fitter, H.N., Pandey, A.B., Patel, D.D., and Mistry, J.M. (2014). A review on approaches for handling bezier curves in cad for manufacturing. *Procedia Engineering*, 97, 1155–1166. doi: https://doi.org/10.1016/j.proeng.2014.12.394. "12th Global Congress on Manufacturing and Management" GCMM - 2014.

Fragapane, G., de Koster, R., Sgarbossa, F., and Strandhagen, J.O. (2021). Planning and control of autonomous mobile robots for intralogistics: Literature review and research agenda. *European Journal of Operational Research*, 294(2), 405–426. doi: https://doi.org/10.1016/j.ejor.2021.01.019.

Ghilardelli, F., Gabriele, L., and Piazzi, A. (2014). Path Generation Using $\eta 4$-Splines for a Truck and Trailer Vehicle. *IEEE Transactions on Automation Science and Engineering*, 11(1), 187–203.

Gim, S., Adouane, L., Lee, S., and Dérutin, J.P. (2017). Clothoids Composition Method for Smooth Path Generation of Car-Like Vehicle Navigation. *Journal of Intelligent and Robotic Systems: Theory and Applications*, 88(1), 129–146.

Klančar, G., Seder, M., Blažič, S., Škrjanc, I., and Petrović, I. (2021). Drivable path planning using hybrid search algorithm based on e* and bernstein–bézier motion primitives. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 51(8), 4868–4882. doi: 10.1109/TSMC.2019.2945110.

Klančar, G., Zdešar, A., Blažič, S., and Škrjanc, I. (2017). *Wheeled mobile robotics - From fundamentals towards autonomous systems*. Elsevier, Butterworth-Heinemann.

Lepetič, M., Klančar, G., Škrjanc, I., Matko, D., and Potočnik, B. (2003). Time optimal path planning considering acceleration limits. *Robotics and Autonomous Systems*, 45, 199–210.

Li, Y., Littlefield, Z., and Bekris, K.E. (2016). Asymptotically optimal sampling-based kinodynamic planning. *The International Journal of Robotics Research*, 35(5), 528–564.

Likhachev, M. and Ferguson, D. (2009). Planning long dynamically-feasible maneuvers for autonomous vehicles. *The International Journal of Robotics Research*, 28(8), 933–945.

Loknar, M.B., Blažič, S., and Klančar, G. (2021). Minimum-time velocity profile planning for planar motion considering velocity, acceleration and jerk constraints. *International Journal of Control*, 1–15. doi: 10.1080/00207179.2021.1987526.

Maqsood, S., Abbas, D.M., Miura, K., Majeed, A., and Iqbal, A. (2020). Geometric modeling and applications of generalized blended trigonometric bézier curves with shape parameters. *Advances in Difference Equations*, 2020, 550. doi:10.1186/s13662-020-03001-4.

Meng, W., Li, C., and Liu, Q. (2021). Geometric modeling of c-bézier curve and surface with shape parameters. *Mathematics*, 9(21). doi:10.3390/math9212651.

Tao, W., Daichuan, Y., Weifeng, L., Chenglin, W., and Baigen, C. (2021). A novel integrated path planning algorithm for warehouse agvs. *Chinese Journal of Electronics*, 30(2), 331–338. doi: https://doi.org/10.1049/cje.2021.02.002.

Velenis, E. and Tsiotras, P. (2008). Minimum-time travel for a vehicle with acceleration limits: Theoretical analysis and receding-horizon implementation. *Journal of Optimization Theory and Applications*, 138(2), 275–296. doi:10.1007/s10957-008-9381-7.

Vivaldini, K.C.T., Galdames, J.P.M., Bueno, T.S., Araújo, R.C., Sobral, R.M., Becker, M., and Caurin, G.A.P. (2010). Robotic forklifts for intelligent warehouses: Routing, path planning, and auto-localization. In *2010 IEEE International Conference on Industrial Technology*, 1463–1468. doi:10.1109/ICIT.2010.5472487.

Webb, D.J. and van den Berg, J. (2013). Kinodynamic RRT*: Asymptotically optimal motion planning for robots with linear dynamics. In *2013 IEEE International Conference on Robotics and Automation (ICRA)*, 5054–5061.

Zamfirache, I.A., Precup, R.E., Roman, R.C., and Petriu, E.M. (2022). Reinforcement learning-based control using q-learning and gravitational search algorithm with experimental validation on a nonlinear servo system. *Information Sciences*, 583, 99–120.

Zdešar, A. and Škrjanc, I. (2018). Optimum velocity profile of multiple bernstein-bézier curves subject to constraints for mobile robots. *ACM Trans. Intell. Syst. Technol.*, 9(5). doi:10.1145/3183891.